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Where I Becomes Infinite

Attention Is All You Need

Module 3.2: Transformer Architecture Deep-Dive

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Session Overview

What You Will Learn

- The historical context of the 2017 paper
- Why recurrence and convolution were abandoned
- The self-attention-only design philosophy
- The full original Transformer architecture
- Impact on every modern LLM

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L2 – Intermediate
- **Module:** 3.2 Transformer Deep-Dive
- **Prerequisite:** What is a Transformer?

Learning Objectives

By the End of This Session You Will Be Able To

1. **Explain** the core claim of the Vaswani et al. 2017 paper and why it was revolutionary
2. **Identify** the three mechanisms the paper replaced with self-attention
3. **Describe** the encoder-decoder structure of the original Transformer
4. **Summarise** the key architectural components: multi-head attention, FFN, positional encoding
5. **Connect** the original paper to modern models (GPT-4, Claude, Gemini)

Prerequisites and Connections

You Should Already Know

- What a Transformer is at a high level
- Token embeddings and sequence representation
- Self-attention mechanism and QKV matrices
- Encoder-only vs. decoder-only families

This Deep-Dive Unlocks

- Scaled Dot-Product Attention (next concept)
- Encoder-Decoder Architecture
- Layer Normalisation and Residual Connections
- Full Transformer Implementation

The State of NLP Before 2017

Dominant Sequence Modelling Approaches

- **Recurrent Neural Networks (RNNs):** Process tokens one at a time, passing a hidden state forward — sequential bottleneck, no parallelism
- **LSTMs and GRUs:** Gated variants that partially address vanishing gradients but remain fundamentally sequential
- **Convolutional Sequence Models:** Parallelisable but limited to fixed local receptive fields; struggle with long-range dependencies
- **Attention-augmented RNNs:** Attention added on top of LSTMs (e.g. Bahdanau, 2015) — better, but recurrence still the backbone

Every dominant architecture at the time had a structural limitation that made scaling difficult.

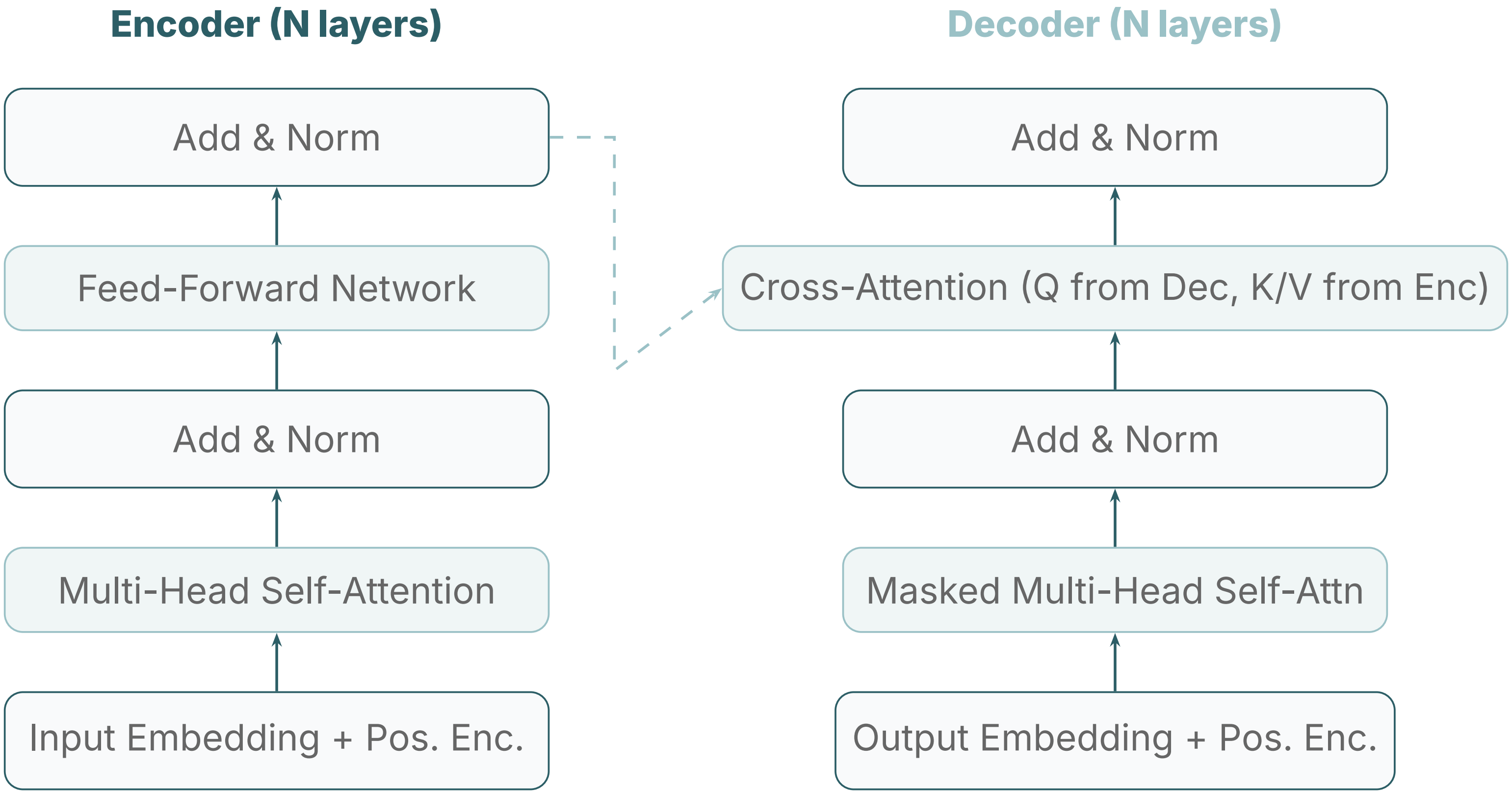
The Central Claim of the Paper

“Attention Is All You Need” — Vaswani et al., NeurIPS 2017

- Recurrence and convolution are not necessary for sequence transduction
- Attention mechanisms alone are sufficient to model dependencies between any two positions in a sequence, regardless of their distance
- The resulting architecture is more parallelisable and reaches higher translation quality than all prior approaches
- Training time on 8 GPUs: 3.5 days for the base model (WMT En-De)

This was a paradigm shift: attention went from being an add-on to being the entire mechanism.

Original Transformer Architecture



Key Architectural Choices in the Original Paper

Design Decisions That Defined Modern Transformers

- $N = 6$ **encoder and decoder layers:** Stacking layers builds progressively more abstract representations
- $d_{\text{model}} = 512$, $h = 8$ **heads:** Each head operates in $d_k = 64$ dimensions; outputs concatenated to d_{model}
- **FFN inner dimension = 2048:** $4 \times d_{\text{model}}$, the ratio maintained in most LLMs today
- **Sinusoidal positional encoding:** No learned positional embeddings; sine/cosine functions of varying frequencies
- **Label smoothing** ($\epsilon = 0.1$) and **dropout** (0.1) for regularisation during training

D

$H_1 \ldots H_8$

$T_1 \ldots T_k$

$8 \times 64 = d_{\text{model}}$

Three Uses of Attention in the Original Transformer

Each Serves a Different Purpose

- **Encoder Self-Attention:** Each encoder token attends to all other encoder tokens — builds full bidirectional context of the source sequence
- **Decoder Masked Self-Attention:** Each decoder token attends only to prior decoder positions — causal masking prevents seeing the future
- **Encoder-Decoder Cross-Attention:** Decoder queries (Q) attend to encoder keys and values (K, V) — the alignment mechanism for seq2seq tasks

These three variants are all mathematically identical (scaled dot-product attention) but differ in what they are allowed to attend to.

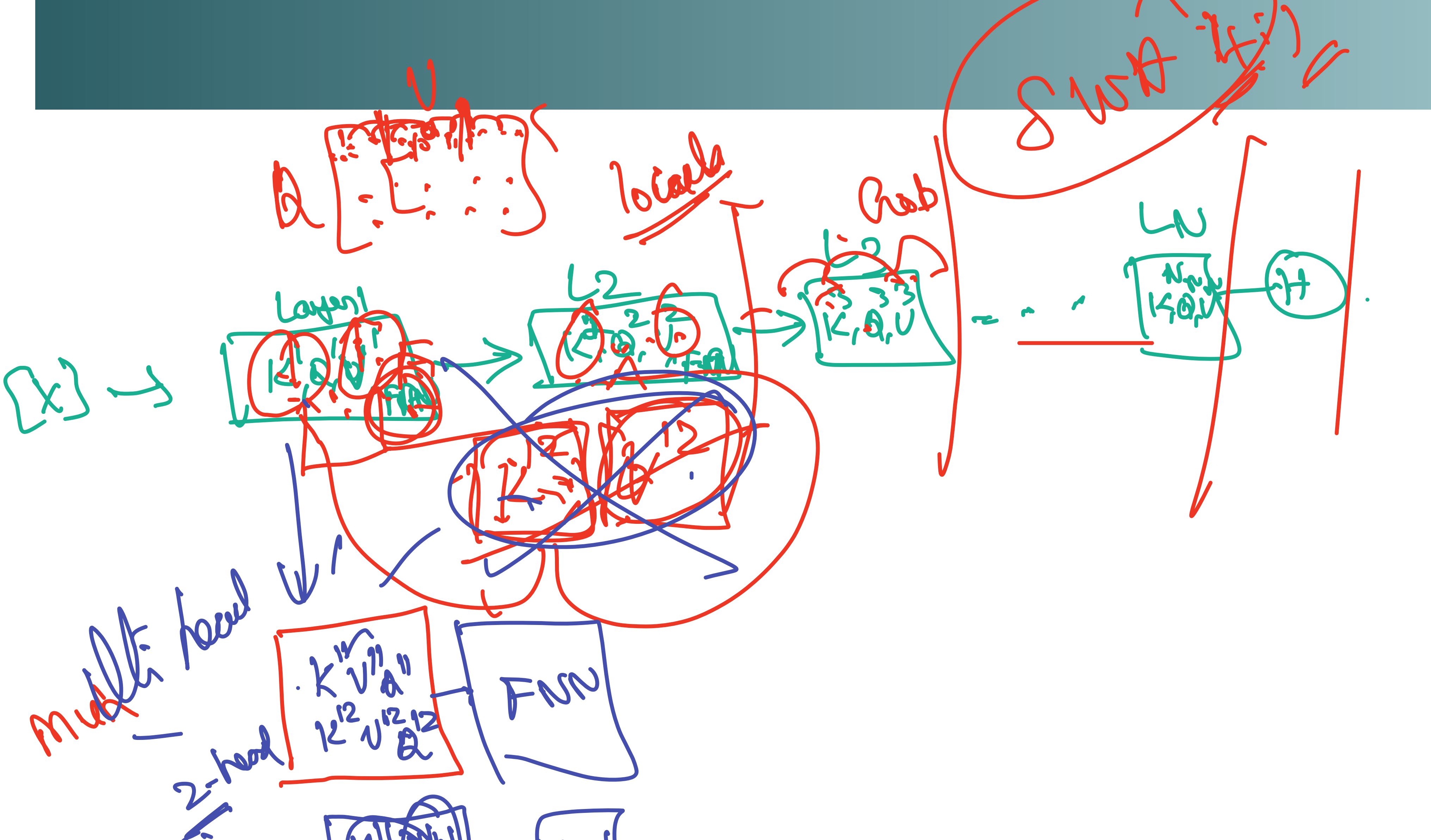
Parallelism: The Practical Revolution

RNN Training (Sequential)

- Token t requires output from token $t-1$
- Cannot be parallelised across time steps
- GPU utilisation: low
- Long sequences: very slow to train

Transformer Training (Parallel)

- All attention scores computed simultaneously
- Full matrix operations on GPU/TPU
- GPU utilisation: very high
- Scales to billions of tokens efficiently



Q
V
locals
Glob
SWA
[X]
Layer1
K, Q, V
FFN
L2
L3
LN
multi head
2-head
FNN

What Has Changed Since 2017

Evolution of the Original Architecture

- **Pre-norm vs. Post-norm:** Original used Post-LN (Add then Norm); modern models (GPT, Llama) use Pre-LN (Norm before attention) for training stability
- **Sinusoidal to RoPE/ALiBi:** Fixed sinusoidal encodings replaced by Rotary Positional Embeddings (RoPE) for better length generalisation
- **Multi-Head to Grouped-Query:** MHA replaced by GQA in Llama 2/3, Mistral to reduce KV-cache memory at inference
- **Softmax to RMSNorm:** LayerNorm replaced by RMSNorm (simpler, faster)
- **GELU / SwiGLU activations:** Replace ReLU in the FFN for better performance

Transformer Families Born from the Original Paper

Encoder-Only

- BERT, RoBERTa
- Bidirectional context
- Classification, NER, QA

Decoder-Only

- GPT-4, Claude, Llama
- Autoregressive generation
- Text generation, chat

Encoder-Decoder

- T5, BART, mT5
- Original structure
- Translation, summarisation

GenAI Application: Neural Machine Translation

The Original Use Case That Motivated the Paper

- **Google Translate:** Powered by Transformer-based models since 2017. WMT English-German BLEU score improved from 26 (ConvS2S) to 28.4 (Transformer base)
- **DeepL:** Uses custom Transformer variants trained on curated parallel corpora; rated highest-quality by professional translators in blind tests
- **Meta SeamlessM4T:** Massively multilingual model handling 100+ languages with the same encoder-decoder Transformer backbone
- **Real-time subtitling:** YouTube, Zoom, and Teams use Transformer ASR + translation pipelines for live multilingual captions

Machine translation was the testbed that proved the Transformer was ready for production.

GenAI Application: Large Language Models

Every Major LLM Is a Transformer Variant

- **GPT-4 (OpenAI):** Decoder-only Transformer, estimated 1T+ parameters; powers ChatGPT with over 100M weekly active users
- **Claude 3.5 Sonnet (Anthropic):** Constitutional AI-aligned Transformer excelling at long-context tasks and code generation (200K context window)
- **Gemini 1.5 Pro (Google):** Multimodal Transformer with a 1M token context window; integrates text, image, audio, and video understanding
- **Llama 3.3 (Meta):** Open-weight decoder-only Transformer enabling on-premises deployment, fine-tuning, and local inference

The 2017 paper's architecture, refined over 8 years, is now the backbone of the AI industry.

GenAI Application: Code Generation

Transformers Writing Software

- **GitHub Copilot:** Based on OpenAI Codex (fine-tuned GPT); generates code completions, docstrings, and tests; used by 1.3M+ developers
- **Google Gemini Code Assist:** Integrated into VS Code, IntelliJ, and Cloud Shell; supports 20+ languages with context-aware completions
- **Amazon Q Developer:** Transformer-based coding assistant embedded in AWS tooling for cloud-native development workflows
- **DeepSeek Coder:** Open-source Transformer trained on 2T code tokens; competitive with GPT-4 on HumanEval benchmark

Code is just another sequence — the Transformer's parallel attention works equally well on tokens from any language.

GenAI Application: Multimodal Transformers

Extending Attention Beyond Text

- **Vision Transformer (ViT):** Treats image patches as tokens and applies standard Transformer encoder to achieve state-of-the-art image classification
- **DALL-E 3 (OpenAI):** Uses Transformer-based CLIP for text encoding and a diffusion decoder for pixel generation from text prompts
- **Whisper (OpenAI):** Encoder-decoder Transformer converting audio spectrograms to text; near-human accuracy across 99 languages
- **GPT-4V / Gemini Vision:** Add visual token embeddings to the Transformer, enabling image + text reasoning in a unified sequence

Common Misconceptions

Clearing Up Confusion About the Paper and Architecture

- **“The paper introduced attention”:** Attention was introduced by Bahdanau et al. (2015) for RNNs. The 2017 paper showed attention *alone* is sufficient.
- **“All modern LLMs are encoder-decoder”:** Most production LLMs (GPT-4, Claude, Llama) are decoder-only. The original paper’s encoder-decoder structure is used mainly for seq2seq.
- **“The Transformer is a black box”:** Every component has a well-understood mathematical role. Interpretability tools like attention visualisation and probing reveal what each layer learns.
- **“Bigger is always better”:** Efficient models like Mistral 7B outperform much larger models on many tasks through architecture improvements, not just parameter count.

The Lasting Impact of the 2017 Paper

Why “Attention Is All You Need” Is a Landmark Work

- One of the most cited papers in computer science history (100K+ citations)
- Established self-attention as the default sequence modelling primitive
- Enabled the pre-train / fine-tune paradigm that defines modern NLP
- Created the architectural foundation for GPT, BERT, T5, and every successor model
- Demonstrated that a single architecture can dominate multiple tasks (translation, summarisation, classification) when scaled

Every LLM in production today is a direct descendant of the ideas in this eight-page paper.

Summary

Attention Is All You Need — Key Takeaways

1. Vaswani et al. (2017) replaced RNNs and CNNs with pure self-attention, enabling full parallelisation of sequence processing
2. The original architecture uses $N{=}6$ encoder and decoder layers, 8-head attention, and $d_{\text{model}}{=}512$
3. Three attention variants serve different roles: encoder self-attention, masked decoder self-attention, and encoder-decoder cross-attention
4. The design has since evolved: Pre-LN, RoPE, GQA, RMSNorm, and SwiGLU are now standard in production LLMs
5. Every major modern model — GPT-4, Claude, Gemini, Llama — descends directly from this architecture

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What This Concept Unlocks

Next Concepts in Module 3.2

- Scaled Dot-Product Attention (deep dive)
- Encoder-Decoder Architecture
- Decoder-Only Architecture
- Layer Normalisation
- Feed-Forward Networks in Transformers

Downstream in the Course

- Full Transformer Implementation (coding)
- Pre-training and Fine-tuning Strategies
- Flash Attention and KV-Cache
- LoRA and Efficient Fine-Tuning

Thank You

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